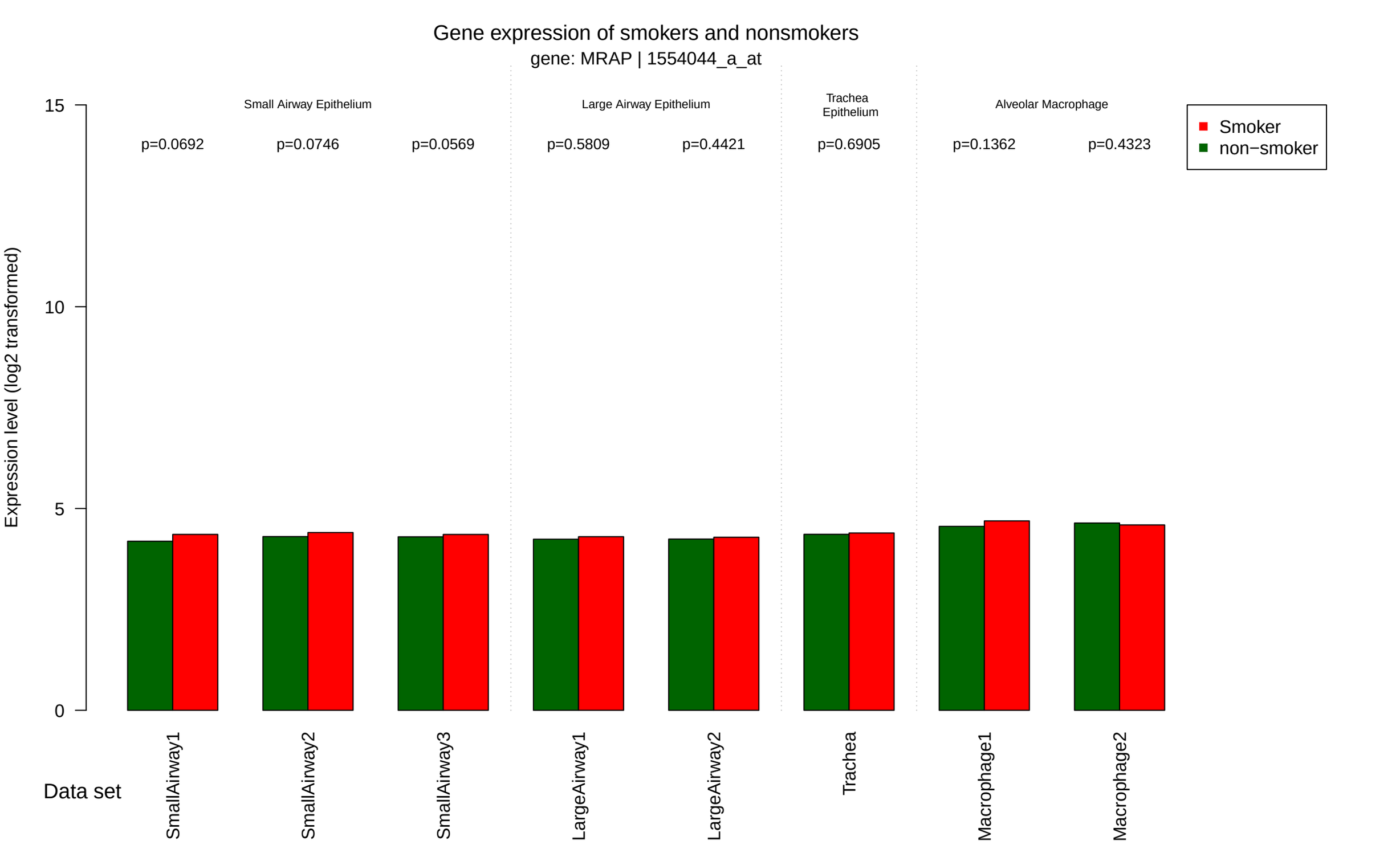
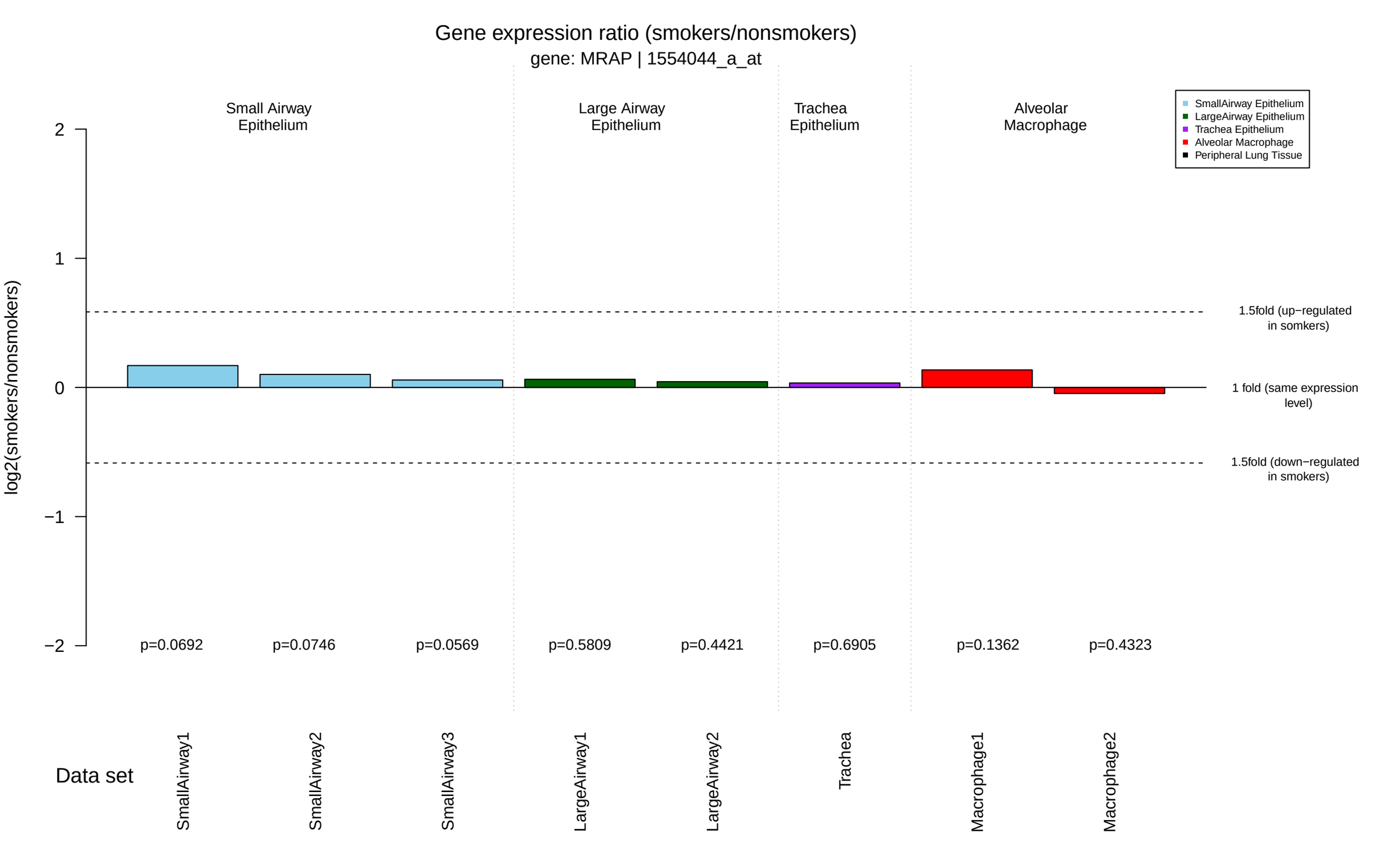
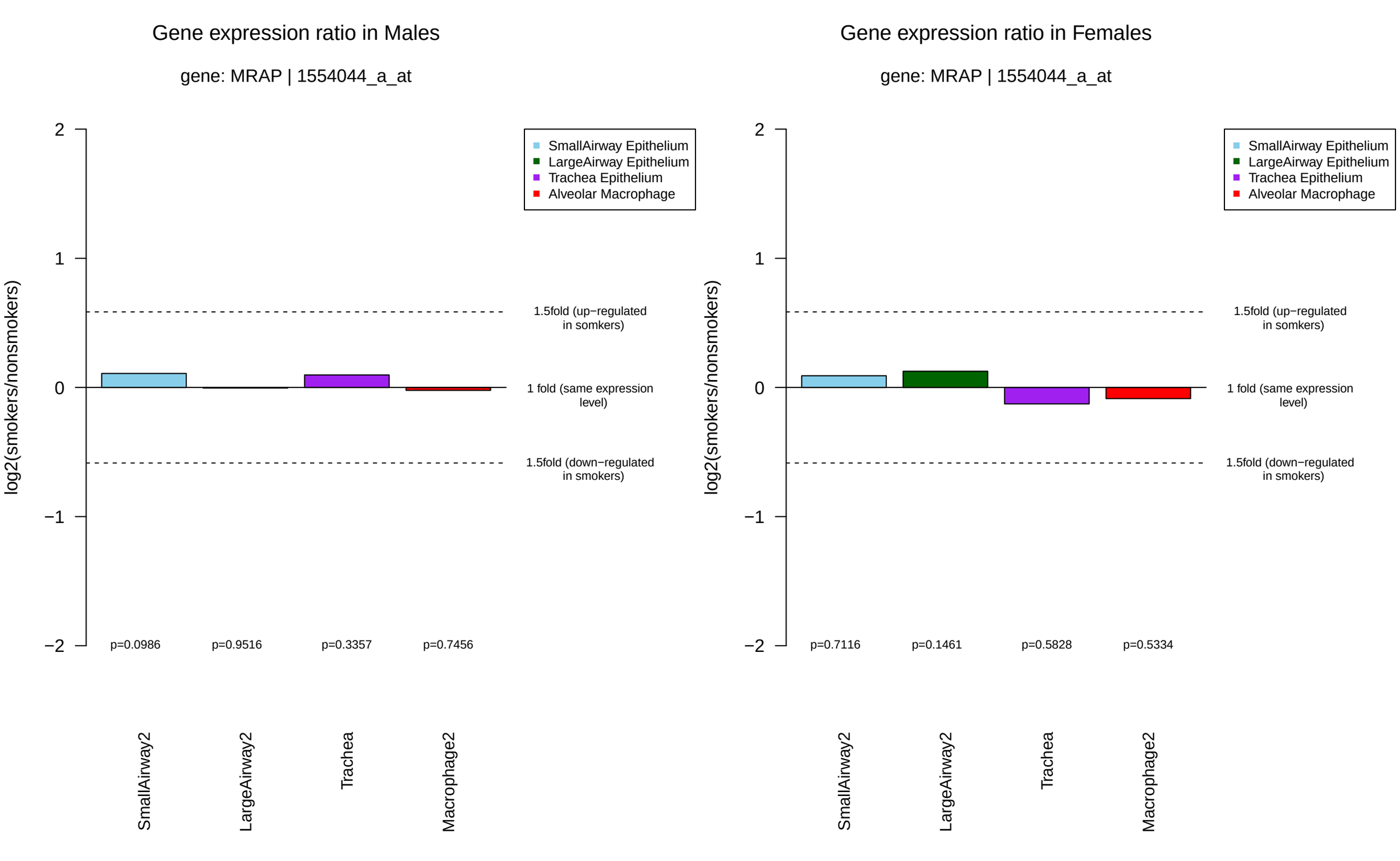
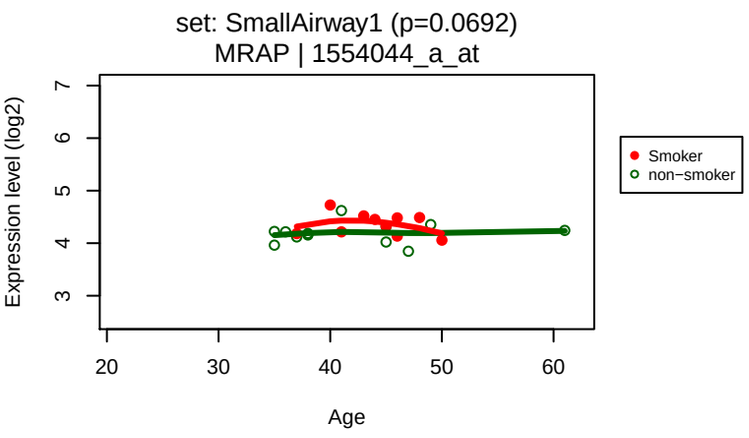
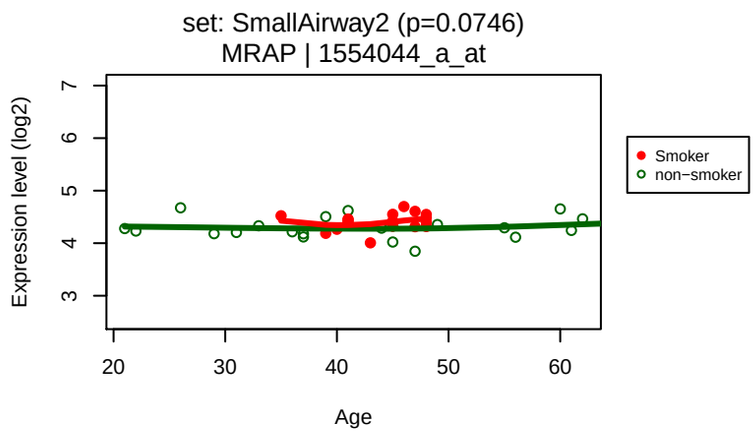
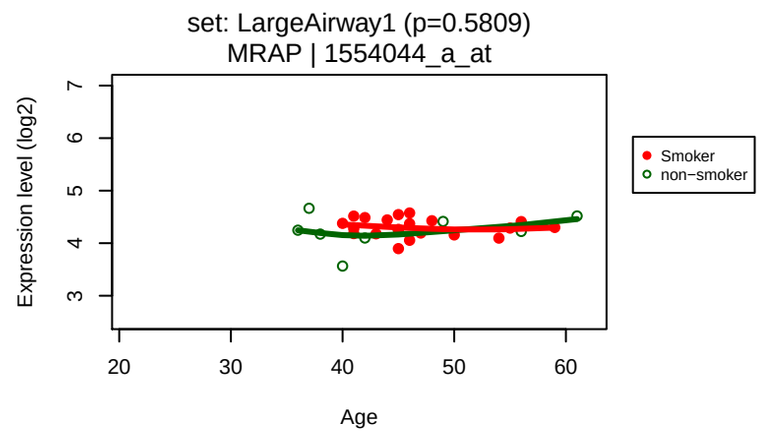

Gene expression of smokers and nonsmokers
gene: MRAP | 1554044_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0692
p=0.0746
p=0.0569
p=0.5809
p=0.4421
p=0.6905
p=0.1362
p=0.4323
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

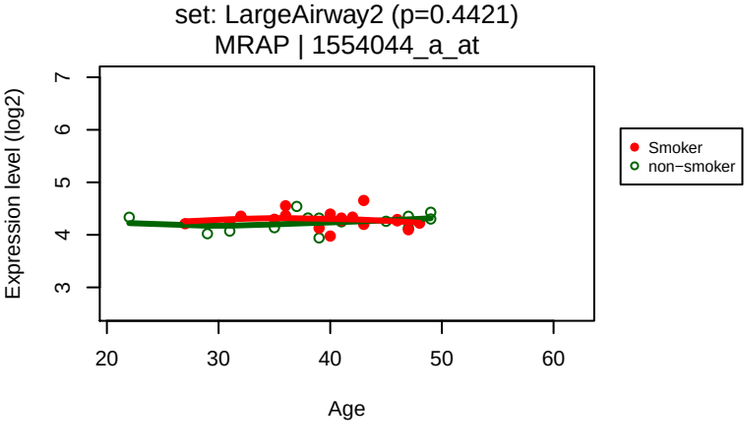

Gene expression ratio (smokers/nonsmokers)
gene: MRAP | 1554044_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0692
p=0.0746
p=0.0569
p=0.5809
p=0.4421
p=0.6905
p=0.1362
p=0.4323
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

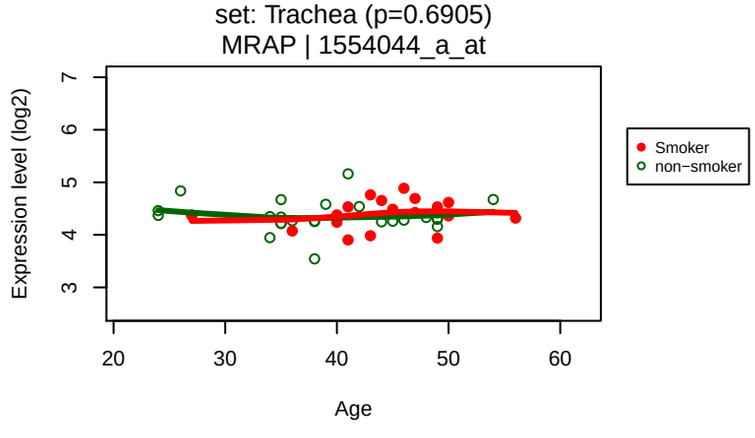

Gene expression ratio in Males
gene: MRAP | 1554044_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0986
p=0.9516
p=0.3357
p=0.7456
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRAP | 1554044_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7116
p=0.1461
p=0.5828
p=0.5334
SmallAirway2
LargeAirway2
Trachea
Macrophage2

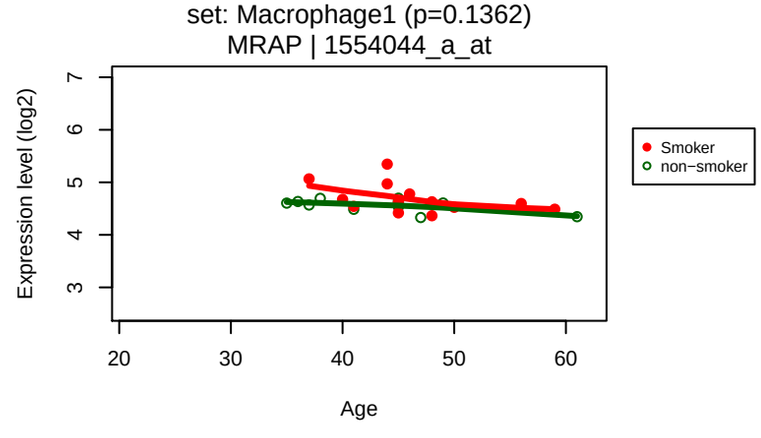

set: SmallAirway1 (p=0.0692)
MRAP | 1554044_a_at
Expression level (log2)
Age
Smoker
non−smoker

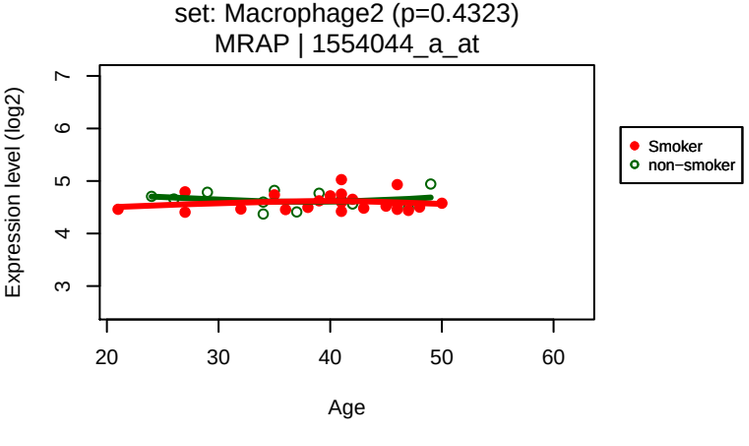

set: SmallAirway2 (p=0.0746)
MRAP | 1554044_a_at
Expression level (log2)
Age
Smoker
non−smoker

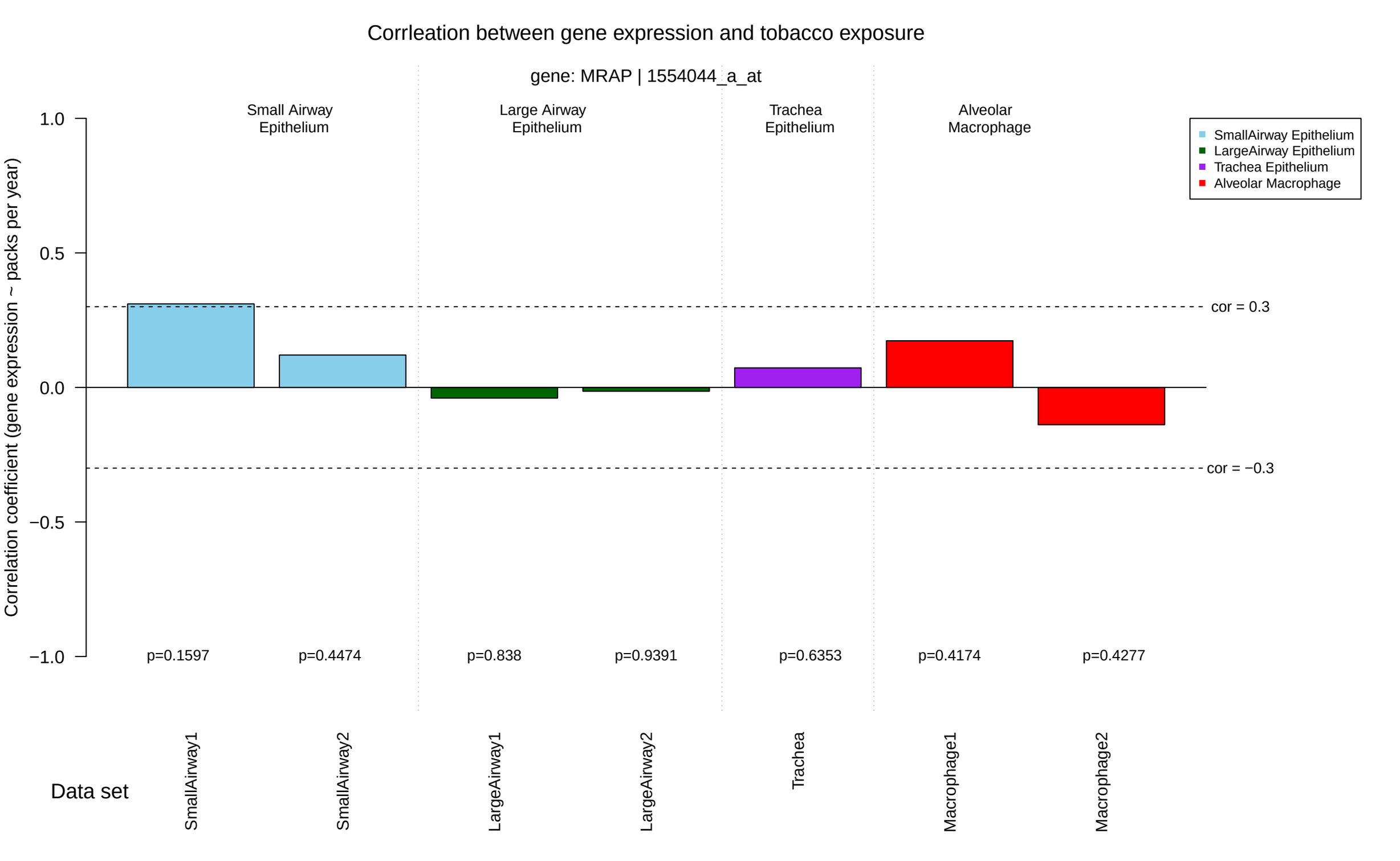

set: LargeAirway1 (p=0.5809)
MRAP | 1554044_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4421)
MRAP | 1554044_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6905)
MRAP | 1554044_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1362)
MRAP | 1554044_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4323)
MRAP | 1554044_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRAP | 1554044_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1597
p=0.4474
p=0.838
p=0.9391
p=0.6353
p=0.4174
p=0.4277
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2